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STA506

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Homework 4

Question 1: Derive the log-logistic density function

$$F(x) = \frac{1}{1 + \left(\frac{x}{\alpha}\right)^{-\beta}}, \quad x > 0, \quad \alpha > 0, \quad \beta > 0$$

Step 1: Rewrite into simpler form

$$\begin{aligned} \left(\frac{x}{\alpha}\right)^{-\beta} &= \left(\frac{\alpha}{x}\right)^{\beta} \\ F(x) &= \frac{1}{1 + \left(\frac{\alpha}{x}\right)^{\beta}} = [1 + \alpha^{\beta} x^{-\beta}]^{-1} \end{aligned}$$

Step 2: Differentiate the CDF to get the PDF

Let

$$g(x) = 1 + \alpha^{\beta} x^{-\beta}$$

Then

$$\begin{aligned} F(x) &= g(x)^{-1} \\ f(x) &= F'(x) = -g(x)^{-2} g'(x) \\ g'(x) &= \alpha^{\beta} \cdot (-\beta) x^{-(\beta+1)} = -\beta \alpha^{\beta} x^{-(\beta+1)} \end{aligned}$$

Substitute into $f(x)$:

$$f(x) = -[1 + \alpha^{\beta} x^{-\beta}]^{-2} (-\beta \alpha^{\beta} x^{-(\beta+1)}) = \frac{\beta \alpha^{\beta} x^{-(\beta+1)}}{[1 + \alpha^{\beta} x^{-\beta}]^2}$$

Step 3: Express in standard log-logistic form

Rewrite the denominator:

$$1 + \alpha^{\beta} x^{-\beta} = 1 + \left(\frac{\alpha}{x}\right)^{\beta} = \frac{x^{\beta} + \alpha^{\beta}}{x^{\beta}}$$

So

$$\left[1 + \left(\frac{\alpha}{x}\right)^\beta\right]^2 = \frac{(x^\beta + \alpha^\beta)^2}{x^{2\beta}}$$

Then

$$\begin{aligned} f(x) &= \frac{\beta \alpha^\beta x^{-(\beta+1)}}{(x^\beta + \alpha^\beta)^2 / x^{2\beta}} \\ &= \beta \alpha^\beta x^{-(\beta+1)} \cdot \frac{x^{2\beta}}{(x^\beta + \alpha^\beta)^2} \\ &= \frac{\beta \alpha^\beta x^{\beta-1}}{(x^\beta + \alpha^\beta)^2} \end{aligned}$$

Divide numerator and denominator by $\alpha^{2\beta}$

$$f(x) = \frac{\left(\frac{\beta}{\alpha}\right) \left(\frac{x}{\alpha}\right)^{\beta-1}}{\left[1 + (x/\alpha)^\beta\right]^2}, x > 0$$

Question 2: Distribution of Recovery Time from a Surgery

a.

Step 1: Population Moments

For a log-logistic distribution with parameters α, β :

$$E[X] = \alpha \frac{\pi/\beta}{\sin(\pi/\beta)}, \quad \beta > 1$$

$$Var(X) = \alpha^2 \left(\frac{2\pi/\beta}{\sin(2\pi/\beta)} - \left[\frac{\pi/\beta}{\sin(\pi/\beta)} \right]^2 \right), \quad \beta > 2$$

Define:

$$A(\beta) = \frac{\pi/\beta}{\sin(\pi/\beta)}$$

$$B(\beta) = \frac{2\pi/\beta}{\sin(2\pi/\beta)} - A(\beta)^2$$

Then

$$E[X] = \alpha A(\beta), \quad Var(X) = \alpha^2 B(\beta)$$

Step 2: Sample Moments

Used this code to determine s^2 and $\bar{x}$

```
## {r}
x <- c(8.23, 12.74, 14.83, 16.61, 18.16, 19.55, 20.80, 21.94, 23.00, 23.98,
      24.89, 25.75, 26.56, 27.34, 28.08, 28.79, 29.48, 30.15, 30.81, 31.45,
      32.08, 32.70, 33.31, 33.92, 34.53, 35.13, 35.73, 36.33, 36.93, 37.53,
      38.14, 38.75, 39.37, 40.00, 40.64, 41.29, 41.95, 42.63, 43.33, 44.05,
      44.79, 45.56, 46.36, 47.20, 48.08, 49.02, 50.03, 51.12, 52.32, 53.65)

n <- length(x)
xbar <- mean(x)
s2 <- var(x) # sample variance (uses n-1)

xbar
s2
##
```

Console

```
> xbar
[1] 34.1922
> s2
[1] 122.1819
> |
```

Compute the ratio:

$$r = \frac{s^2}{\bar{x}^2} = \frac{122.1819}{(34.1922)^2} \approx 0.10451$$

Step 3: Equate Population and Sample Moments

$$\bar{x} = \alpha A(\beta)$$

$$s^2 = \alpha^2 B(\beta)$$

$$\frac{s^2}{\bar{x}^2} = \frac{B(\beta)}{A(\beta)^2}$$

$$0.10451 = \frac{B(\beta)}{A(\beta)^2}$$

Step 4: Numerical Solution

```

```{r}
Compute the ratio s^2 / xbar^2 from sample moments
r <- s2 / (xbar^2)

Define A(beta) = (pi/beta) / sin(pi/beta)
This corresponds to the theoretical expression in E[X] = alpha A(beta)
A <- function(beta) (pi/beta) / sin(pi/beta)

Define B(beta) from the theoretical variance expression
Var(X) = alpha^2 B(beta)
B <- function(beta) (2*pi/beta) / sin(2*pi/beta) - A(beta)^2

Define nonlinear function g(beta) = r - B(beta)/A(beta)^2
We solve g(beta)=0 to eliminate alpha and estimate beta
g <- function(beta) r - (B(beta) / (A(beta)^2))

beta must be greater than 2 for the variance to exist
Use uniroot() to numerically solve g(beta)=0
beta_hat <- uniroot(g, lower = 2.05, upper = 30)$root

Once beta_hat is found, solve for alpha_hat using xbar = alpha A(beta)
alpha_hat <- xbar / A(beta_hat)

Display the method-of-moments estimates
beta_hat
alpha_hat
```

```

Console

```

> beta_hat
[1] 5.952531
> alpha_hat
[1] 32.62681
>

```

$$\hat{\alpha} = 32.6268$$

$$\hat{\beta} = 5.9525$$

b.

```

####{r}
set.seed(506)          # ensure reproducibility
B_reps <- 2000          # number of bootstrap resamples
n <- length(x)

alpha_star <- numeric(B_reps)
beta_star <- numeric(B_reps)

for (b in 1:B_reps) {

  # Generate bootstrap sample (resample with replacement)
  x_star <- sample(x, size = n, replace = TRUE)

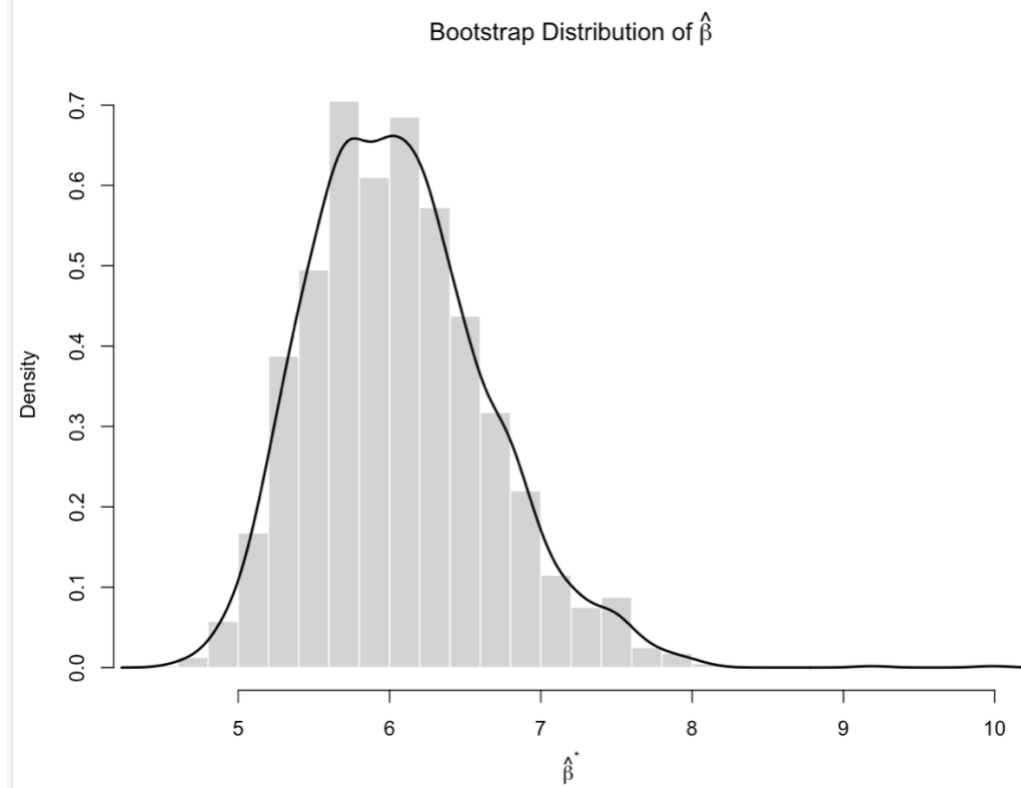
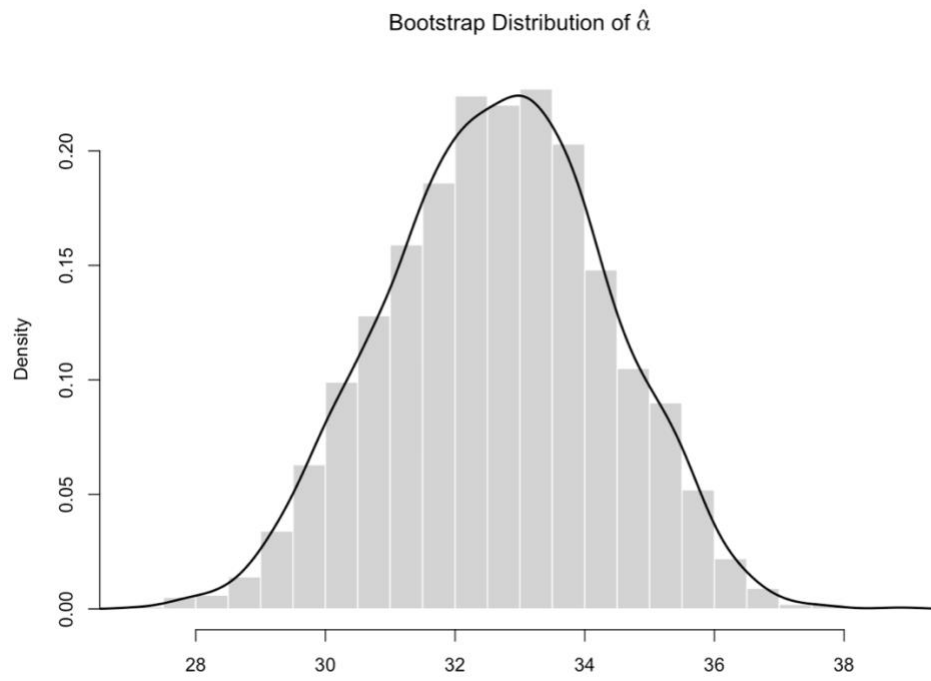
  # Recompute MoM estimates on bootstrap sample
  est <- mom_loglogistic(x_star)

  alpha_star[b] <- est["alpha_hat"]
  beta_star[b] <- est["beta_hat"]
}

# Plot bootstrap distribution of alpha_hat
hist(alpha_star, breaks = 30, probability = TRUE,
      main = expression("Bootstrap Distribution of " * hat(alpha)),
      xlab = expression(hat(alpha)^"*"), border = "white")
lines(density(alpha_star), lwd = 2)

# Plot bootstrap distribution of beta_hat
hist(beta_star, breaks = 30, probability = TRUE,
      main = expression("Bootstrap Distribution of " * hat(beta)),
      xlab = expression(hat(beta)^"*"), border = "white")
lines(density(beta_star), lwd = 2)
####

```



The bootstrap distribution for $\hat{\alpha}$ appears fairly symmetric and centered around approximately 32.6. Most of the resampled estimates fall within a relatively narrow range, and the density curve does not show noticeable skewness. This suggests that the scale parameter estimate is fairly stable across bootstrap samples and is not highly sensitive to small changes in the data.

The bootstrap distribution for $\hat{\beta}$, however, shows more spread and a slightly longer right tail. While most estimates are concentrated near 6, some bootstrap samples produce larger values, which creates the visible right skew. This is reasonable because $\hat{\beta}$ is obtained by solving a nonlinear equation, making it more sensitive to variation in the resampled data, especially when larger recovery times appear more frequently.

Overall, the bootstrap results indicate that the estimate of α is more stable than the estimate of β , with β showing greater variability across repeated samples.